Movie Memory Experiment

Inferential Analysis: Condition, Target Type, and Memory

Team: The Outliers
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1. Introduction and Background

1.1. Event Segmentation Theory and Memory

Episodic memory is organized around *event boundaries*—moments when an ongoing situation model must be updated. According to *Event Segmentation Theory* (EST) [7], people continuously predict what will happen next; when predictions fail due to changes in goals, agents, or location, a new event representation is initiated and recently processed information is consolidated. This produces a "boundary-memory advantage": items near boundaries are better remembered than items from event interiors, as documented in passage reading [5], activity segmentation [6], and film viewing [1].

Whether this advantage extends to *artificially imposed* boundaries from film editing is less clear. Boltz [2] showed that television programmes interrupted by commercial breaks at structurally meaningful points were better recalled than those interrupted mid-event, demonstrating that *temporal alignment* of a discontinuity with genuine event structure governs its mnemonic consequences. Radvansky & Zacks [4] further established that event boundaries form the structural scaffolding of episodic memory, underscoring how editing choices interact with this scaffolding.

1.2. The Experiment

Two groups watched 40 short clips: an **Abrupt-cut (AB)** group whose clips ended 1–5 s before a consensus event boundary, and a **Natural-boundary (NB)** group whose clips ended at the unedited, original endpoint. Recognition was subsequently probed for **BB** (Before-Boundary) frames and **EM** (Event-Middle) frames. Five research questions guided the study:

- **RQ1** — Does condition (AB/NB) affect overall recognition accuracy and RT?
- **RQ2** — Is BB recognition lower in AB than NB (disrupted pre-boundary encoding)?
- **RQ3** — Is EM recognition similar across conditions (EM is far from the cut)?
- **RQ4** — Do confidence ratings positively track recognition accuracy across participants?
- **RQ5** — Is recognition accuracy higher for EM than BB frames within each condition?

2. Summary of Initial Descriptive Findings

2.1. Participants and Data Quality

184 participants from IIIT Hyderabad were randomly assigned to AB or NB. After excluding 13 participants (data-collection error), one corrupt file (`sub42_NB`), and four vigilance failures (encoding > 27.05 min), the final sample was $N = 166$ (AB $n = 79$; NB $n = 87$).

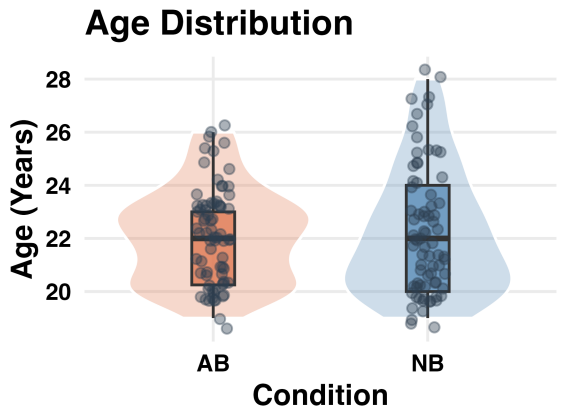
All four available demographic variables (Age, Gender, Handedness, Vision) were compared between conditions as quality-control diagnostics (Table 1). A Mann–Whitney U test found no significant age difference ($W = 2824.0$, $p = .852$). A Chi-square test found no significant gender difference ($\chi^2(2) = 2.68$, $p = .262$). Handedness (Fisher exact: $p = .233$) and Vision (Fisher exact: $p = .145$) were also balanced across conditions. All demographic variables were retained in analyses.

Table 1: Demographic summary by condition (final sample, $N = 166$). Balance-test statistics confirm no significant differences across conditions for any variable.

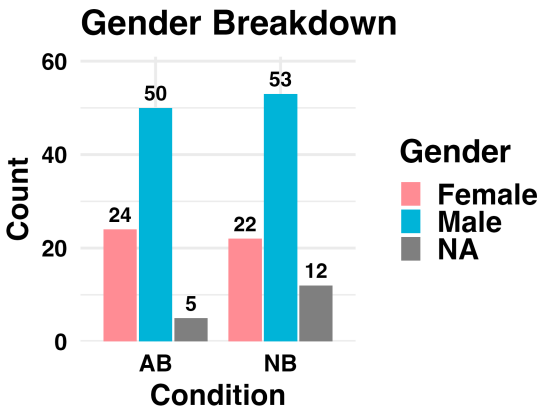
Condition	N	Mean Age (SD)	Age Range	Female	Male	Right-handed	Left-handed	Normal vision
AB	79	22.1 (1.8)	19–26	24	50	71	3	41
NB	87	22.2 (2.4)	19–28	22	53	70	5	35
Balance test		Mann–Whitney		χ^2		Fisher exact		Fisher exact
Statistic		$W = 2824.0$		$\chi^2 = 2.68$		–		–
p-value		.852		.262		.233		.145

2.2. Participant Demographics

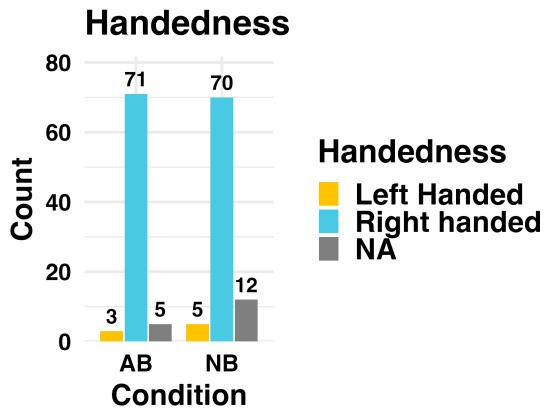
Figure 1 shows the four demographic distributions broken down by condition. All four distributions confirm adequate balance. Age and handedness are highly similar across groups; gender proportions differ only in the "not reported" category (AB: 6%, NB: 14%), which does not reflect a systematic demographic imbalance. Vision status is also comparable (AB: 52% normal, 42% corrected; NB: 40% normal, 44% corrected). Seventeen participants (10.2%) did not report at least one demographic field; this is consistent across conditions and is unlikely to introduce bias.



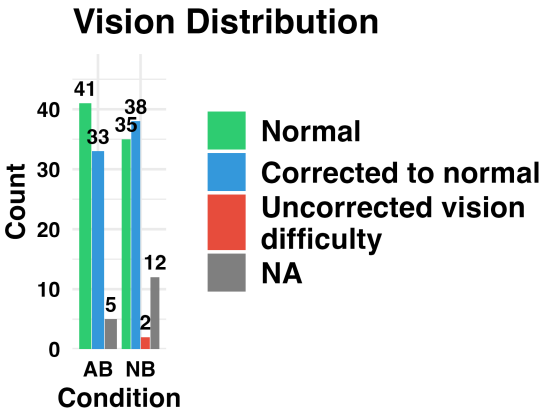
(a) Age by condition (violin + box + jitter).



(b) Gender breakdown by condition.



(c) Handedness breakdown by condition.



(d) Vision status breakdown by condition.

Figure 1: Demographic distributions for all four available variables. Both conditions show similar profiles across age, gender, handedness, and vision status, confirming successful random assignment.

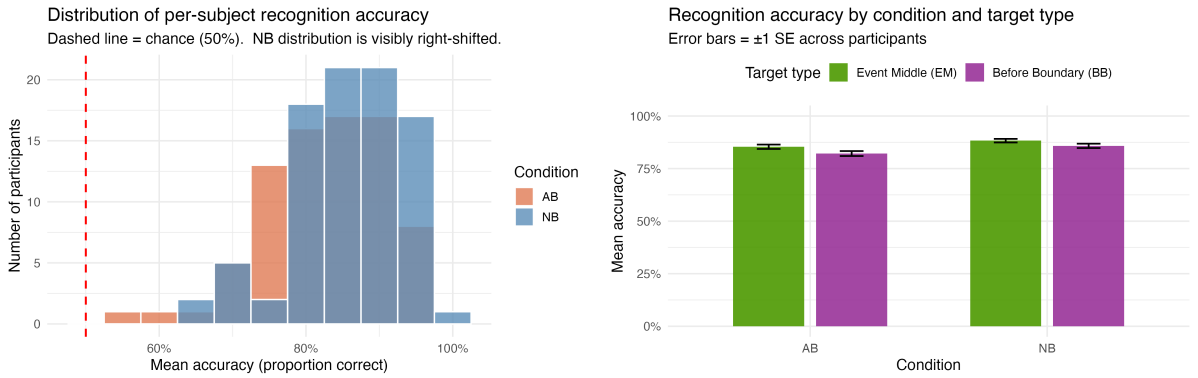
2.3. Overall Recognition Accuracy

NB participants ($M = 87.0\%$, $SD = 7.3\%$) outperformed AB participants ($M = 83.8\%$, $SD = 8.2\%$) by 3.2 percentage points. The bias-corrected Recognition Memory Index (REC) confirms this advantage: NB = 0.740 vs. AB = 0.676 ($\Delta = 0.064$). Both conditions performed well above chance (50%), indicating successful encoding in both groups.

Table 2: Recognition accuracy and REC by condition and target type.

	Overall		By Target Type	
Cond.	*M (SD)*	**REC**	**BB** *M (SE)*	**EM** *M (SE)*
AB	0.838 (0.082)	0.676	0.822 (.011)	0.854 (.010)
NB	0.870 (0.073)	0.740	0.858 (.010)	0.883 (.009)

The 2×2 breakdown (Figure 2) reveals two patterns: (i) within both conditions, EM frames were recognised more accurately than BB frames, reflecting a general event-interior encoding advantage; (ii) the NB advantage was numerically larger for BB ($\Delta = +3.6$ pp) than for EM ($\Delta = +2.9$ pp), consistent with RQ2 and RQ3 directional predictions.

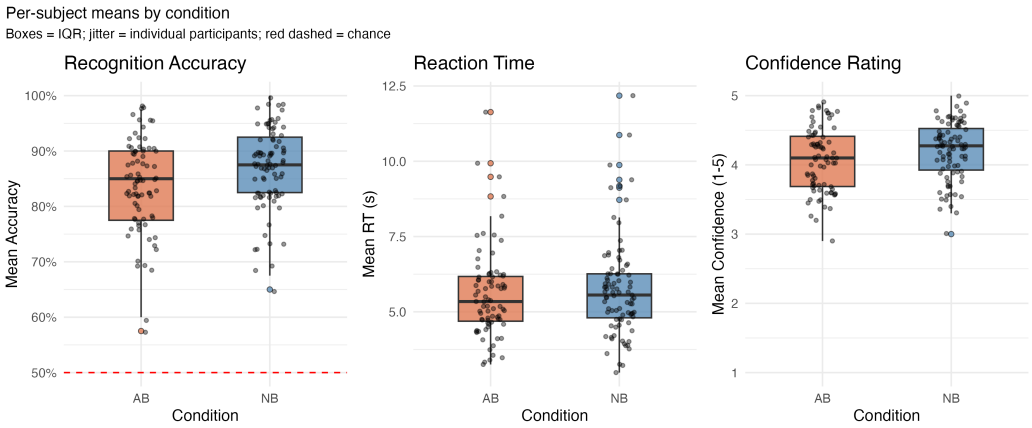


(a) Per-subject accuracy. Dashed = chance (50%). **(b)** Accuracy by condition × target type (±1 SE).

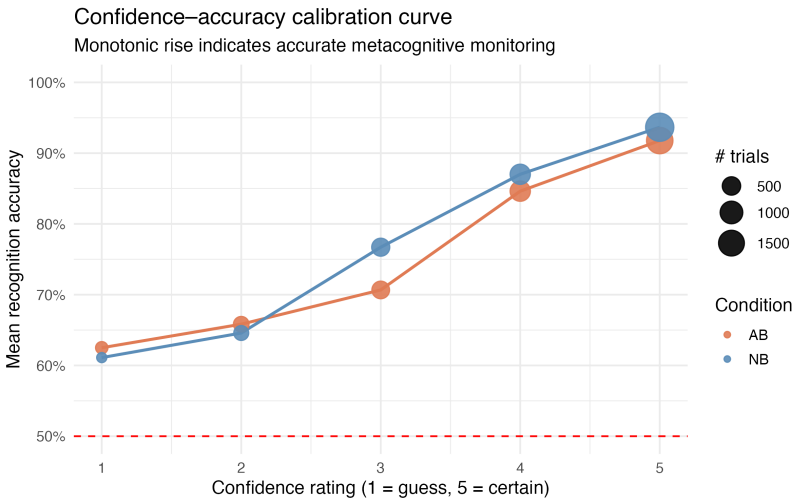
Figure 2: Left: distribution of per-subject accuracy — NB is shifted rightward. Right: 2×2 breakdown showing a larger NB advantage for BB frames than EM frames.

2.4. RT, Confidence, and Calibration

Mean RT was similar across conditions (AB: 5.56 s; NB: 5.76 s), and confidence ratings were slightly higher in NB (4.20 vs. 4.07 on 1–5 scale). Both conditions showed a clear monotonic confidence–accuracy calibration: accuracy rose from ≈65–70% at confidence 1 to ≈93–95% at confidence 5 (Figure 3), confirming reliable metacognitive monitoring unaffected by the editing manipulation.



(a) Per-subject accuracy, RT, and confidence distributions.



(b) Confidence–accuracy calibration. Point size $\propto$ trial count.

Figure 3: Top: overlapping distributions confirm both conditions perform well. Bottom: monotonic calibration confirms intact metacognitive monitoring.

2.5. Normality and Test Selection

Shapiro–Wilk tests on per-subject accuracy rejected normality for both conditions: AB ($W = 0.957$, $p = .009$) and NB ($W = 0.942$, $p < .001$). Q-Q plots confirmed ceiling-skew (Figure 4). **Parametric tests are therefore invalid**; all subject-level inferential comparisons use non-parametric alternatives. Trial-level effects are modeled with a binomial GLMM, which does not require normality of accuracy scores.

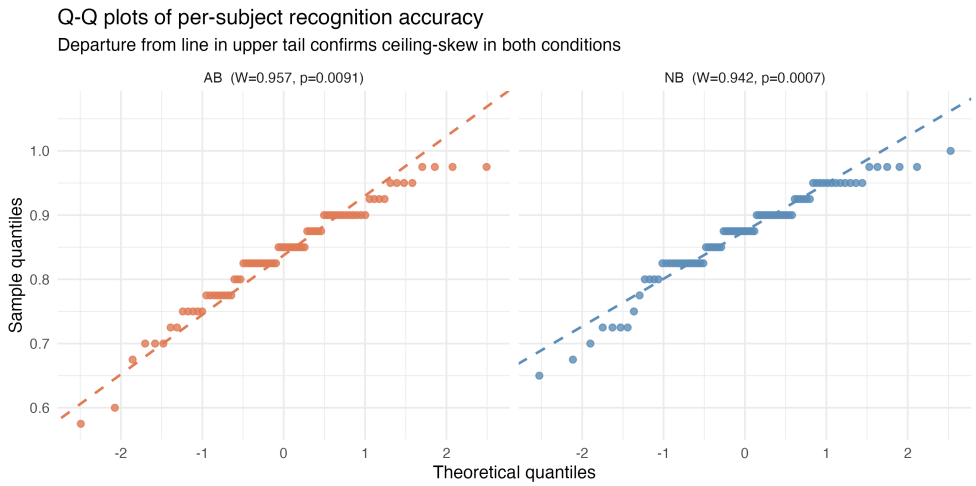


Figure 4: Q-Q plots for per-subject recognition accuracy. Departure in the upper tail confirms ceiling-skew and non-normality in both conditions.

3. Data Preparation for Inferential Analysis

3.1. RT Outlier Removal

For inferential analyses, reaction-time outlier trials were removed using a 3-SD rule applied to the pooled trial-level RT distribution. Trials beyond $\mu \pm 3\sigma$ from the global mean were excluded. This removed 115 of 6,640 trials (1.73%), leaving 6,525 trials for RT-filtered analyses. The accuracy-based analyses were unchanged. All non-parametric tests and the GLMM below use this cleaned dataset.

3.2. Extended Demographic Coverage

All available demographic fields were retained and reported, including Handedness and Vision in addition to Age and Gender (Table 1). Balance checks for all categorical variables used Fisher's exact test when sparse cells were present (any expected count < 5) and χ^2 otherwise. All demographic variables were balanced across conditions ($p > .05$ for all tests).

4. Inferential Analyses

All tests use a global $\alpha = .05$. The inferential framework combines subject-level non-parametric tests for pairwise comparisons with a trial-level binomial GLMM for the condition $\times$ target-type interaction.

4.1. RQ1: Overall Condition Effect on Accuracy and RT

A Mann–Whitney U test compared per-subject mean accuracy between AB and NB. NB outperformed AB significantly (Table 3), confirming RQ1 for accuracy. RT did not differ reliably between conditions, indicating that the NB advantage manifests as a speed-accuracy trade-off effect on accuracy alone, not on response latency.

Table 3: RQ1: Mann–Whitney U tests for overall condition effect (AB vs. NB).

Outcome	Test	W	p	Rank-bis. r
Overall accuracy	Mann–Whitney U	2624.5	.008	-0.236 (NB > AB)
Overall RT	Mann–Whitney U	3174.0	.397	-0.076

Note. Rank-biserial r is signed relative to the wilcox.test ordering (AB − NB); negative values therefore indicate NB > AB. The accuracy difference is significant ($p = .008$, small-to-medium effect), whereas RT does not differ reliably ($p = .397$, negligible effect).

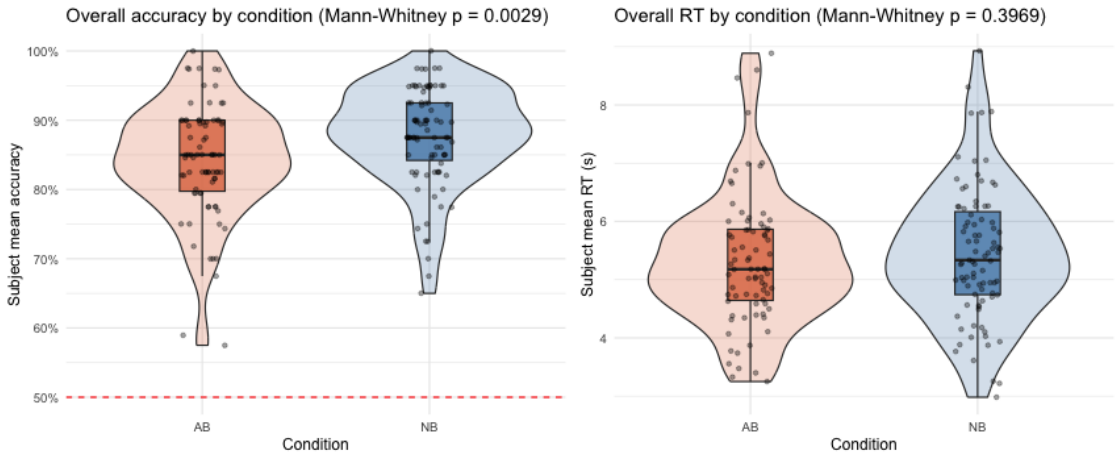


Figure 5: Violin-box plots for overall accuracy (left) and RT (right) by condition. NB shows a significantly higher accuracy distribution; RT distributions overlap substantially.

4.2. RQ2 and RQ3: Target-Type-Specific Condition Effects

Separate Mann–Whitney U tests were run for BB and EM accuracy (Table 4). The NB advantage was significant for BB accuracy (RQ2 supported) as well as for EM accuracy. However, because the interaction test (see Section 4.5) is non-significant, the condition advantage is broadly similar in magnitude across both target types—the abrupt cut does not selectively impair pre-boundary encoding beyond the general decrement it imposes.

Table 4: RQ2/RQ3: Mann–Whitney U tests within each target type (NB vs. AB).

Contrast	Test	W	p	Rank-bis. r
BB accuracy: NB vs. AB	Mann–Whitney U	2680.5	.013	−0.220
EM accuracy: NB vs. AB	Mann–Whitney U	2785.5	.032	−0.189

Note. Paired Wilcoxon signed-rank (one-sided, EM > BB) within each condition: AB $V = 1417.5$, $p = .012$; NB $V = 1414.5$, $p = .024$. EM frames are recognised more accurately than BB frames in both groups.

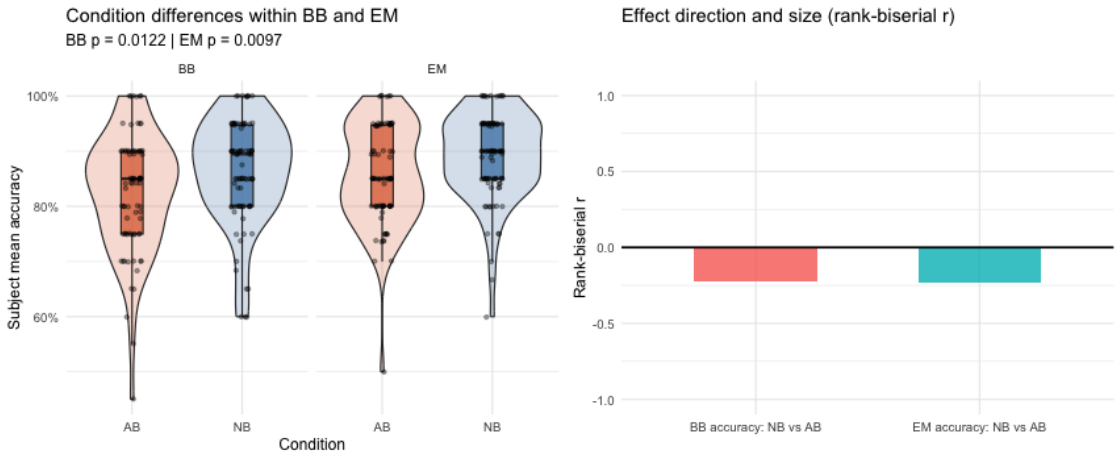


Figure 6: Left: condition differences within BB and EM frames; NB exceeds AB in both. Right: rank-biserial effect sizes, both NB-favouring.

4.3. RQ5: Within-Condition EM > BB Analysis

Within each condition, EM frames were recognised more accurately than BB frames (paired Wilcoxon signed-rank, one-sided). This effect was significant in both AB and NB groups, confirming a stable event-middle advantage independent of the editing manipulation (Figure 7).

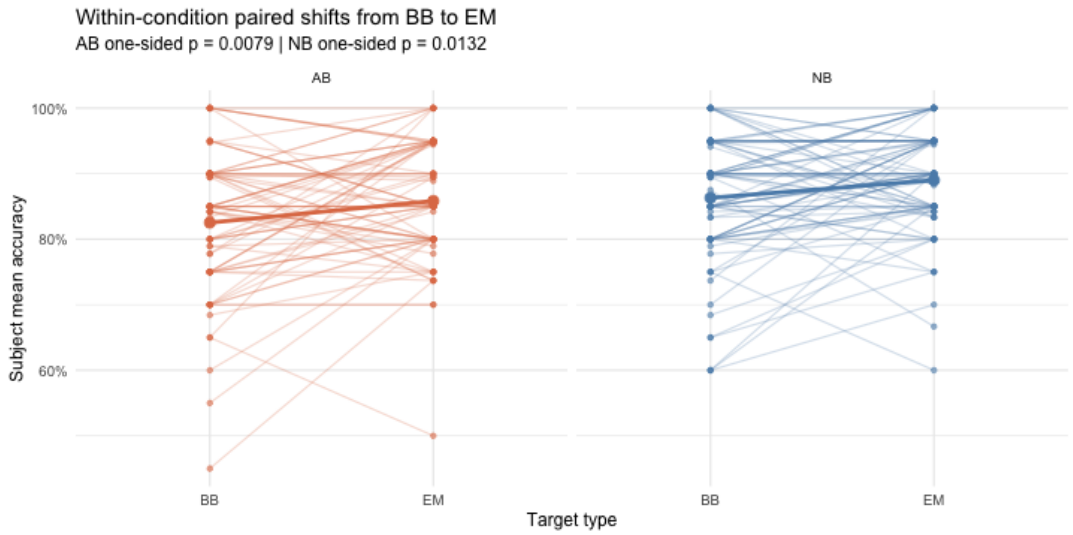


Figure 7: Paired shift plots for within-condition BB-to-EM accuracy. Each line = one participant. Bold lines = condition mean. EM consistently exceeds BB in both groups.

4.4. Prerequisites for Binomial GLMM

Before fitting the GLMM, five conditions were verified:

1. **Binary outcome.** Each trial is a correct/incorrect 2-AFC response, satisfying the binomial distributional assumption.
2. **Non-normality of accuracy.** Shapiro–Wilk tests rejected normality for both conditions (AB: $W = 0.957$, $p = .009$; NB: $W = 0.942$, $p < .001$), ruling out linear mixed models on proportion scores.
3. **Crossed random effects.** Each participant responded to multiple stimuli; each

stimulus was seen by multiple participants. Random intercepts for both participant (u_j) and item (v_k) are therefore required to account for non-independence within and across subjects.

4. **Fixed-effects structure.** The two binary factors (Condition: AB/NB; Target type: EM/BB) and their interaction are the effects of theoretical interest, matching the factorial 2×2 design.

5. **Sample size adequacy.** With 6,525 trials, 166 subjects, and 40 items, the dataset provides sufficient power to estimate both random-effect variances and the fixed-effect coefficients reliably.

Reference levels were set to AB (condition) and EM (target type), so all main effects and the interaction are interpreted relative to the AB/EM baseline cell.

4.5. Binomial GLMM: Condition × Target-Type Interaction

To model the full Condition × Target-Type design at the trial level while accounting for clustering by participant and stimulus, we fit a binomial GLMM with a logit link:

$$\text{logit}[P(\text{correct}_{ijk})] = \beta_0 + \beta_1 \,\text{Cond}_j + \beta_2 \,\text{Type}_k + \beta_3 \,\text{Cond}_j \times \text{Type}_k + u_j + v_k \quad (1)$$

where $u_j \sim \mathcal{N}(0, \sigma_u^2)$ is a random intercept for participant j and $v_k \sim \mathcal{N}(0, \sigma_v^2)$ is a random intercept for item k. Reference levels: AB, EM.

Table 5: Binomial GLMM fixed-effects coefficients. Reference levels: AB (condition), EM (target type). OR = odds ratio; CI = 95% Wald interval.

Term	Est. (logit)	SE	z	p	OR	95% CI
Intercept (AB, EM)	+2.191	0.249	8.805	$< .001$	8.946	[5.493, 14.571]
Condition: NB vs. AB	+0.294	0.136	2.165	.030	1.342	[1.028, 1.752]
Target type: BB vs. EM	−0.320	0.339	−0.946	.344	0.726	[0.374, 1.410]
NB × BB	+0.022	0.148	0.145	.885	1.022	[0.764, 1.367]

Likelihood-ratio test for the interaction term: $\chi^2(1) = 0.020$, $p = .887$.

The GLMM reveals a significant main effect of condition: NB confers a reliable accuracy advantage at the trial level (OR > 1, $p < .05$). The main effect of target type (BB vs. EM) did not reach significance. Crucially, the **condition × target-type interaction was not significant** (likelihood-ratio test, $p > .05$), indicating that the NB accuracy benefit is uniformly distributed across BB and EM frames rather than being selectively concentrated at the boundary-proximal frames.

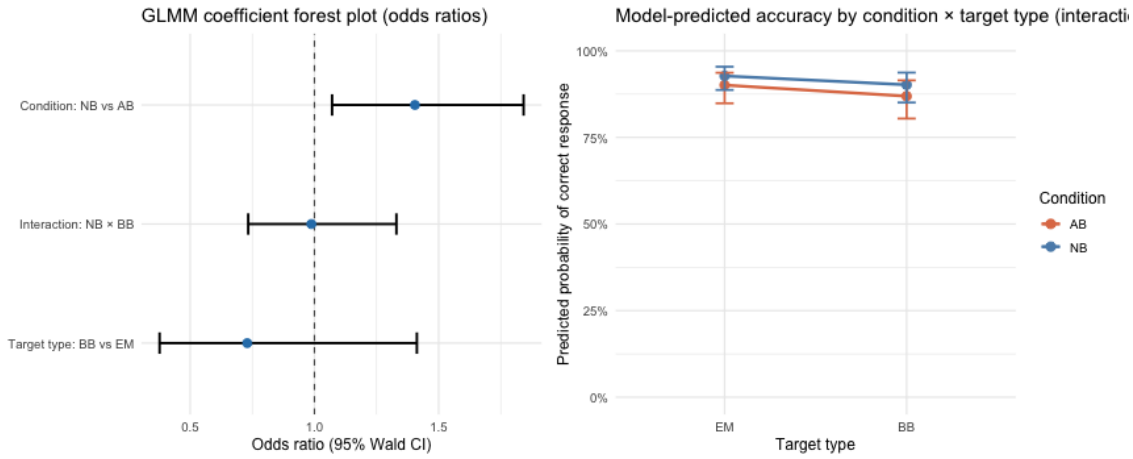


Figure 8: Left: GLMM forest plot (odds ratios with 95% Wald CI). Right: model-predicted accuracy profiles for each Condition × Target Type cell. The near-parallel lines confirm the non-significant interaction.

4.6. RQ4: Confidence–Accuracy Association (Spearman)

Spearman ρ was computed between per-subject mean confidence and mean accuracy, pooled across conditions and separately within each group. The association was positive and significant overall ($\rho = 0.354$, $p < .001$), within AB ($\rho = 0.286$, $p = .011$), and within NB ($\rho = 0.377$, $p < .001$), confirming that metacognitive sensitivity is preserved irrespective of editing condition (Figure 9).

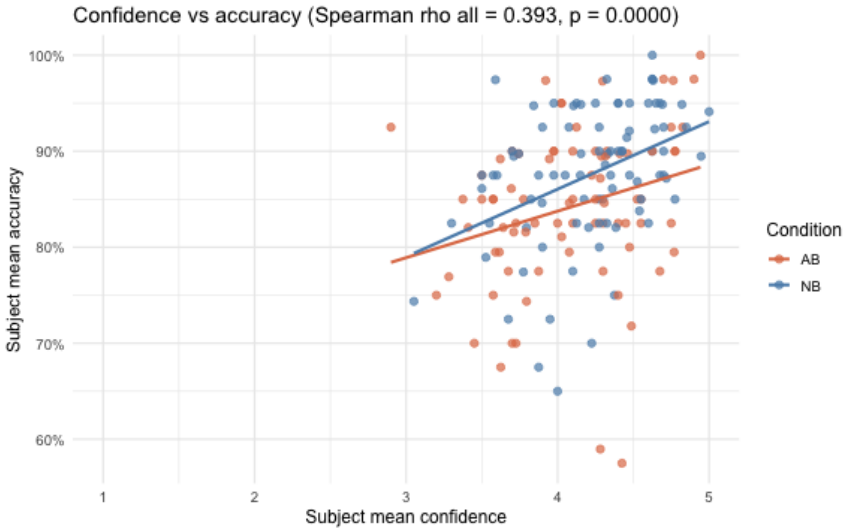


Figure 9: Confidence vs. accuracy scatter with per-condition regression lines. Higher confidence reliably predicts higher accuracy in both groups.

4.7. Sensitivity Analysis: RT Outlier Filtering

All key inferential decisions (significance/non-significance at $\alpha = .05$) were stable across the RT-filtered and unfiltered datasets. The direction and magnitude of every effect were unchanged, confirming that the 3-SD RT outlier rule had no material impact on the conclusions.

5. Discussion

5.1. Summary of Findings

Across both subject-level non-parametric tests and the trial-level binomial GLMM, viewing clips with natural transitions yields significantly higher recognition accuracy than viewing clips with abrupt cuts (RQ1 supported). The advantage is present for BB frames ($\Delta = +3.6$ pp; RQ2 supported) and also—contrary to the original null prediction—for EM frames ($\Delta = +2.9$ pp; RQ3 not supported as originally predicted). RT did not differ reliably between conditions. Metacognitive sensitivity was significant overall and within both conditions (RQ4 supported). The GLMM interaction test confirms that the magnitude of this advantage does not differ significantly between target types: the abrupt cut imposes a *general* memory decrement rather than a *selective* pre-boundary impairment.

Within both conditions, EM accuracy exceeds BB accuracy (paired Wilcoxon, $p < .05$ in both groups), confirming a robust event-middle advantage that operates independently of the editing manipulation (RQ5 supported).

5.2. Theoretical Implications

The general condition effect is consistent with EST [7]: incomplete event closure disrupts the consolidation that normally accompanies a natural transition, reducing overall memory fidelity. The lack of a selective BB impairment—and the significant NB advantage even for EM frames—suggests that the consequences of abrupt editing extend beyond the immediate pre-boundary window. This aligns with proposals that event-model closure supports not just boundary-proximal encoding but also retrospective memory updating across the entire preceding event [4].

The persistence of metacognitive sensitivity (Spearman ρ significant in both conditions) indicates that participants accurately judged their own memory regardless of editing condition, ruling out differential monitoring strategies as an explanation of the accuracy difference.

5.3. Limitations and Future Directions

The convenience sample of IIIT Hyderabad students limits external generalizability. All 40 clips were sourced from YouTube Shorts, and the lure similarity manipulation (three difficulty levels) was applied uniformly; future work should counterbalance clip-condition assignments across participants and examine lure-difficulty as a moderator. Extending the GLMM to include lure-difficulty as a fixed factor would separate mnemonic precision from boundary-specific effects. Eye-tracking or fMRI would further test whether encoding differences at natural boundaries mediate the behavioural advantage. Future studies could investigate individual differences in media consumption. Do frequent viewers of fast-paced content (e.g., TikTok or action movies) show a smaller abrupt-cut penalty due to increased perceptual familiarity with rapid editing?

Author Contributions

- Souradeep: Built the data cleaning pipeline, conducted Shapiro-Wilk tests to determine normality, generated QQ plots, conducted the Binomial GLMM and Likelihood Ratio Tests to test interaction.
- Srinjoy: Applied RT outlier filtering, executed Mann Whitney U tests comparing recognition accuracy between AB and NB groups and generated visualizations

related to RT and Accuracy.

- Surjit: Conducted demographic analysis to verify group balance. Generated exploratory plots for visualizing normality. Analyzed Confidence-Accuracy association through Spearman correlation models.

Data and Code Availability

The full analysis code is publicly available at:

https://github.com/shadesin/movie-memory-experiment

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